Investigating the Impacts of PKM2 Silencing and Hypoxic Stress on Inflammatory Response and Energy Metabolism in Sepsis-Induced Cardiomyopathy

Cellular & Molecular Biology (CELL)

Purpose

This study aims to investigate the role of Pyruvate-Kinase M2 (PKM2) in mitigating inflammatory responses and ATP production, potentially offering insights into a new therapeutic approach for sepsis-induced cardiac dysfunction.

Background

- Sepsis (SIC)
  - Occurs when the body's response to infection becomes dysregulated, causing widespread inflammation and potential organ dysfunction.
- Hypoxia
  - Hypoxia can exacerbate the conditions of SIC due to the body's new reliance on anaerobic glycolysis
- Aerobic glycolysis (Warburg Effect)
  - Proliferation of tumor and cancer cells
  - Cancer cells tend to convert most glucose to lactate regardless of whether oxygen is present
- Pyruvate Kinase M2 (PKM2)
  - A key enzyme involved in aerobic glycolysis
  - Releases pro-inflammatory cytokines
  - Reduction may disrupt normal glycolytic pathways that supply energy to inflammatory cells

H9c2 are clonal cell lines derived from embryonic BD1X rat heart tissue (Used to inhibit septic model)

Previous studies have shown that the PKM2 inhibitor, Shikonin, has been discovered to reduce serum lactate levels and protect mice from lethal endotoxemia. H9c2 cells might have similar results

Activitated by Lipopolysaccharides, or LPS (Used for inducing Sepsis onto cells for cardiovascular research

SiRNA transfection is a method used to inhibit proteins by degrading them before they turn into proteins (Used for inhibiting PKM2)

Figure 1, 2, 3: Background Infographic

Objectives

Research Question: What are the effects of silencing PKM2 expression in a septic and hypoxic model of H9c2 cardiomyocytes on ATP production and inflammatory release?

Hypothesis: If PKM2 is silenced, using siRNA, then it will result in a reduced inflammatory response and reduced production of ATP because PKM2 plays a crucial role in supplying pro-inflammatory cytokines that contribute to the production of ATP.

Controls:

- H9c2 with no treatment (-)
- H9c2 with LPS only (-)
- H9c2 with Hypoxia only (-)
- Hypoxia + siRNA (30nm, 60nm, 120nm) (-)

Methods

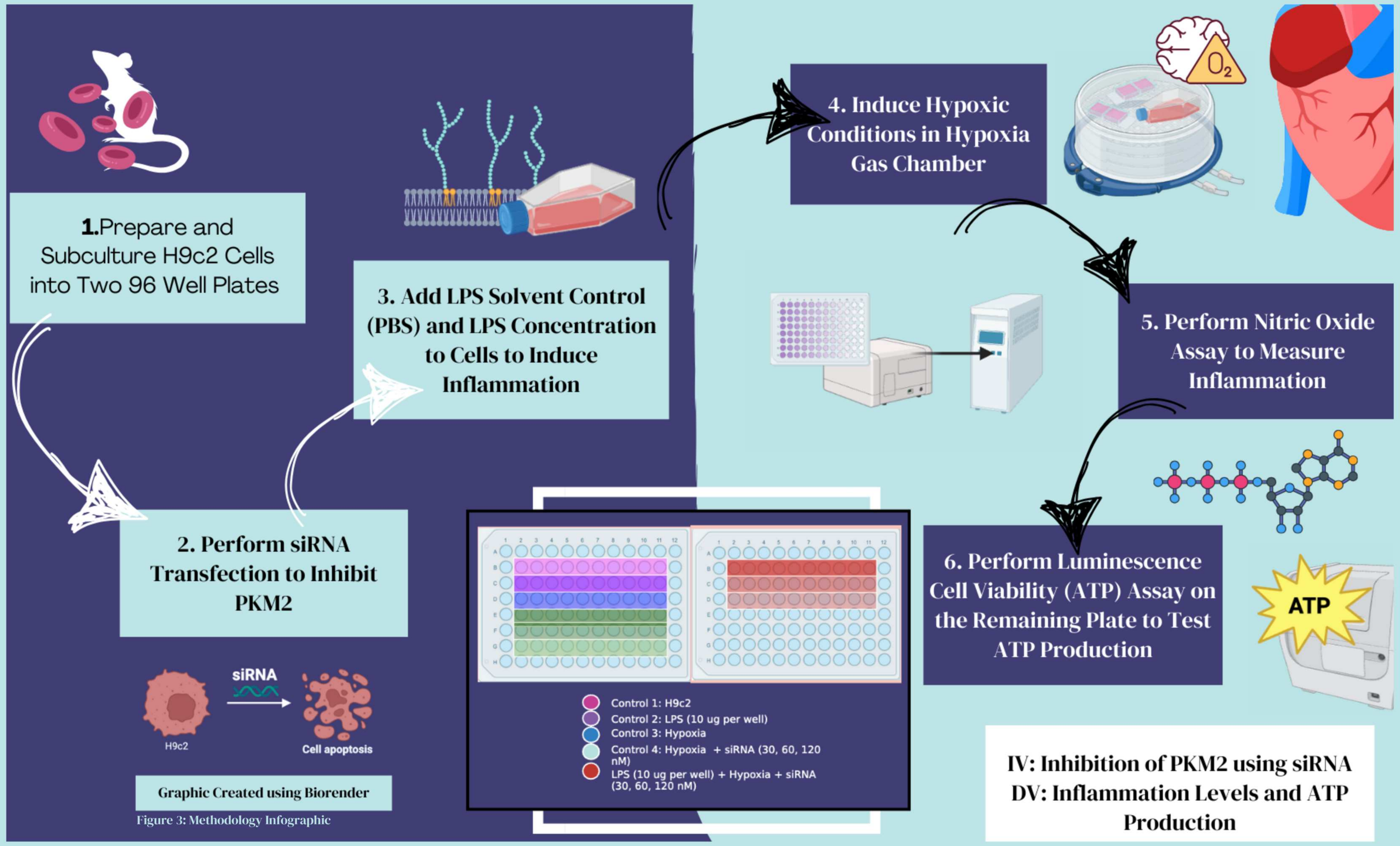

Figure 3: Methodology Infographic

IV: Inhibition of PKM2 using siRNA
DV: Inflammation Levels and ATP Production

Data & Results

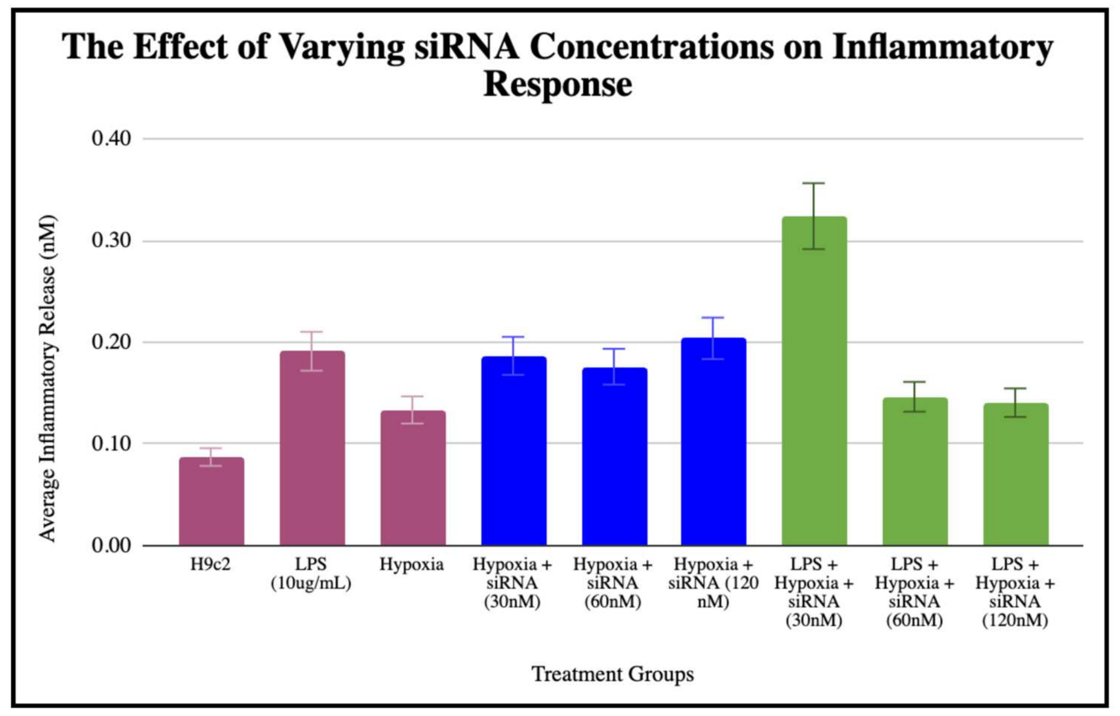

Figure 4: Nitric Oxide Assay Graph (n=10) replicates per group

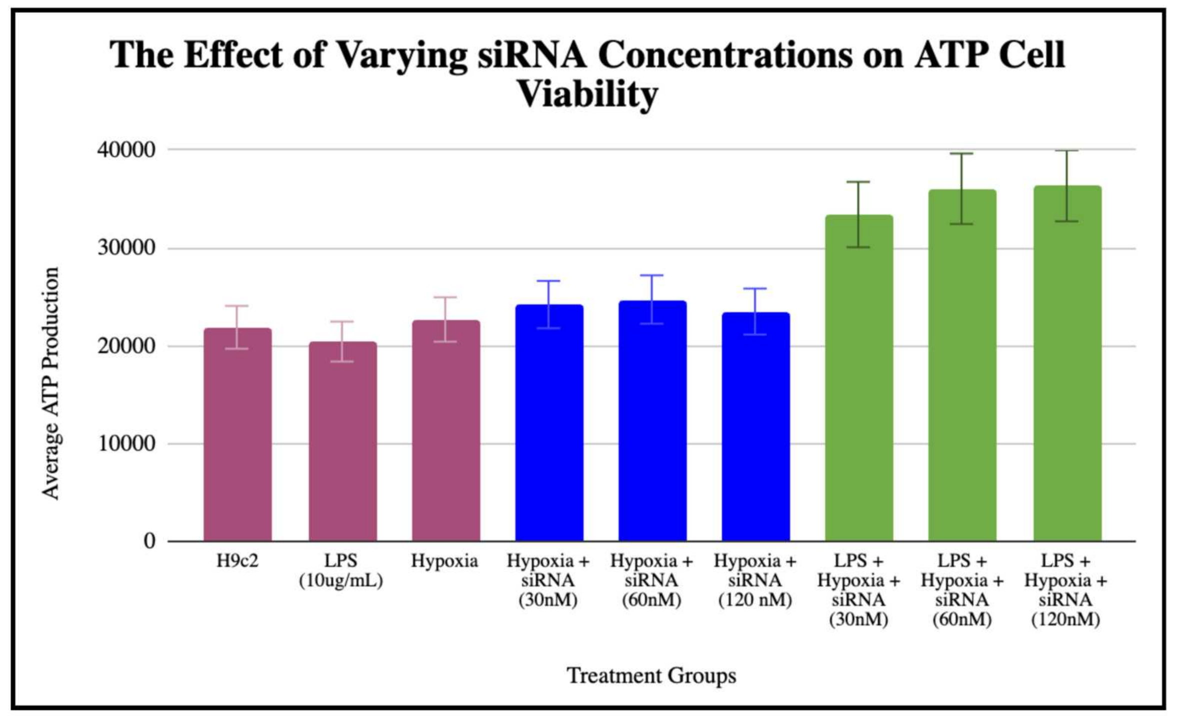

Figure 5: ATP Cell Viability Assay Graph (n=10) replicates per group

Nitric Oxide ANOVA Test Results:

- The averages of the different control group inflammatory release values were taken into consideration a parametric ANOVA Test was run. It resulted in a p-value of 0.0083 which rejects the null hypothesis and indicates that there is a significant difference between differing siRNA concentrations and inflammatory release.

ATP Cell Viability Kruskal-Wallis Test Results:

- A nonparametric Kruskal-Wallis Statistics Test was run on the ATP release from LPS + Hypoxia + siRNA concentrations. It resulted in a p-value of 0.0012 which rejects the null hypothesis and indicates that there is a significant difference between differing siRNA concentrations and ATP production. However, the graph clearly displays that there was an increase in ATP production with our variables.

Discussion

Since the p-value from both statistical tests was $p < 0.05$, the results from this experiment support the hypothesis that the reduction of PKM2 expression will affect the inflammatory release and ATP production in H9c2 cells after being stimulated with LPS and Hypoxia. The data does support a drastic decrease in inflammatory release with increasing concentrations of siRNA, it does not support a decrease in ATP production. Figure 5 displays a slight decrease in ATP production with just Hypoxia + siRNA, however there was a sudden spike in ATP production once the LPS was added to the cells. This could be due to various factors such as LPS boosting glycolysis or triggering other energy-producing pathways as a quick stress response, especially since PKM2, a vital enzyme, was knocked down. Based on the graphs, siRNA concentrations and inflammation have a negative correlation, and do not appear to be strong because the results of cell viability varied greatly among the three trials. There is less variability in the cell viability of cells added to a 60 nM and 120 nM siRNA concentration compared to 30 nM, suggesting that 30 nM siRNA is an optimal amount to elicit in the inflammatory response without decreasing cell viability greatly. Additional research needs to be conducted to test if inhibition of PKM2 truly protects the cells from inflammation rather than just reducing the pro-inflammatory cytokine supply, therefore PKM2 cannot be determined as a therapeutic target against sepsis just yet.

Future Work

It is crucial to continue this research to further delve into PKM2's role in a septic setting under hypoxic stress, but also to discover potential new targets. Because cancer cells and septic undergo the same process that is aerobic glycolysis, it would be important to test the processes of inhibiting other key proteins involved in glycolysis and dysregulating the glycolytic pathways of cancer cells moving forward. The significant results are a step closer to identifying severe cases of systemic inflammatory response syndrome (SIRS) and will assist future studies in identifying key targets against it. Ultimately, the goal would be to develop feasible treatments that improve outcomes for patients with sepsis-induced cardiac dysfunction under hypoxic stress.

References

Alquraishi, M., Puckett, D. L., Alani, D. S., Humidat, A. S., Frankel, V. D., Donohoe, D. R., Whelan, J., & Bettaieb, A. (2019). Pyruvate Kinase M2: A Simple Molecule with Complex Functions. Free Radical Biology & Medicine, 143, 176–192. https://doi.org/10.1016/j.freeradbiomed.2019.08.007

Boccellino, M., Galasso, G., Ambrosio, P., Stiuso, P., Lama, S., Di Zazzo, E., Schiavon, S., Vecchio, D., D'ambrosio, L., Quagliuolo, L., Feola, A., Frati, G., & Di Domenico, M. (2021). H9c2 Cardiomyocytes under Hypoxic Stress: Biological Effects Mediated by Sentinel Downstream Targets. Oxidative Medicine and Cellular Kongevity, 2021, 6874146. https://doi.org/10.1155/2021/6874146

L'Heureux, M., Sternberg, M., Brath, L., Turlington, J., & Kashiouris, M. G. (2020). Sepsis-Induced Cardiomyopathy: a Comprehensive Review. Current cardiology reports, 22(5), 35. https://doi.org/10.1007/s11886-020-01277-2

Liu, Z., Le, Y., Chen, H., Zhu, J., & Lu, D. (2021). Role of PKM2-Mediated Immunometabolic Reprogramming on dDvelopment of Cytokine Storm. Frontiers in Immunology, 12. https://doi.org/10.3389/fimmu.2021.748573

Qiu, Zhen, He, Yuhong, Ming, Hao, Lei, Shaoqing, Leng, Yan, Xia, Zhong-yuan. (2019). Lipopolysaccharide (LPS) Aggravates High Glucose- and Hypoxia/Reoxygenation-Induced Injury through Activating ROS-Dependent NLRP3 Inflammasome-Mediated Pyroptosis in H9C2 Cardiomyocytes. Journal of Diabetes Research, 8151836, 12 pages, 2019. https://doi.org/10.1155/2019/8151836